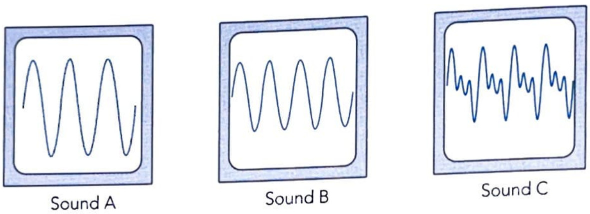
TRIAL TEST 5: WAVES

Time allowed:	**60 minutes**	**Section One** – Short response	18 marks
Total marks:	**60**	**Section Two** – Problem solving	30 marks
		Section Three – Comprehension	12 marks

SECTION ONE – SHORT RESPONSE (18 MARKS)

1. Three different sounds A, B and C were played into a microphone connected to a cathode ray oscilloscope (CRO). Their individual traces are shown below. Assume similar settings of the CRO for each sound.



Describe the difference/s that a listener would notice between:

(a) Sounds A and B

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(b) Sounds B and C

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[2 marks]

2. Under similar conditions sound waves of all frequencies travel at the same speed. From your knowledge of the nature of sound explain clearly why this statement must be correct.

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[2 marks]

3. A loud sound travelling in cool air, encounters a region of warm air. Indicate briefly how each of the following may be affected as the sound passes through to the warm air and beyond.

(a) Velocity ______________________________

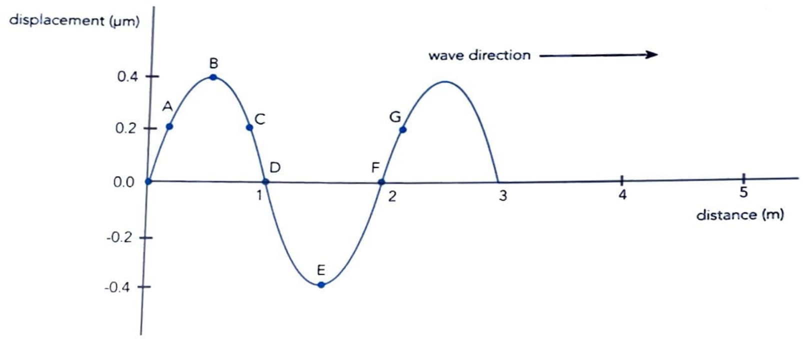
(b) Direction ______________________________

(c) Wavelength ______________________________

(d) Frequency ______________________________

[4 marks]

4. The graph below shows the displacement of air molecules from their mean position at an instant in time, $t = 0$.



(a) From the graph determine the following:

(i) amplitude ______________

(ii) wavelength ______________

(iii) two points in phase ______________

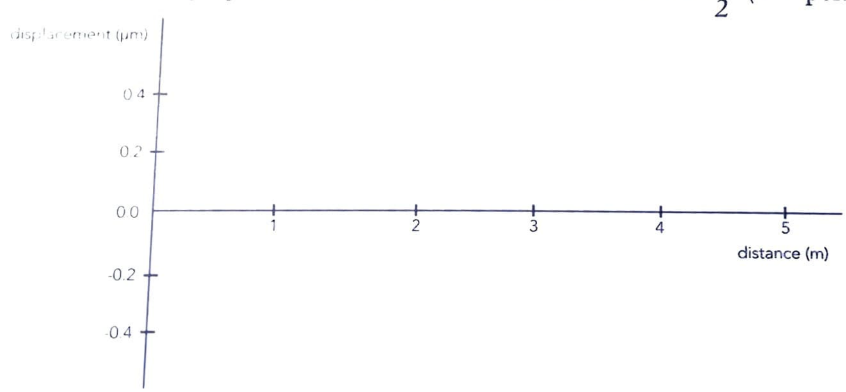
(iv) two stationary points ______________

(v) a point moving away from its mean position ______________

(vi) two points moving with greatest speed ______________

[3 marks]

(b) Redraw the graph above to show how it will look at $t = \frac{T}{2}$ (T = period).



[3 marks]

5. Echo sounding devices are often used to determine the depth of oceans and rivers. These devices emit short pulses of ultrasound (about 50 kHz) towards the ocean floor and then detect the reflected sound.

(a) Explain how it is possible to measure ocean depth in this way.

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[1 mark]

(b) Why are such high frequency sounds used for this purpose?

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[1 mark]

(c) If the delay time between sounding and receiving a signal is 110 ms what is the depth of the water? (You may need to refer to physical data tables.)

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[2 marks]

SECTION TWO – PROBLEM SOLVING (30 MARKS)

6. A loudspeaker emits a sound of 1.00 kHz towards a microphone as shown. The microphone is 2.00 m from the speaker. Assume the speed of sound is 340 ms^{-1}.

(a) Determine the wavelength of the sound being emitted.

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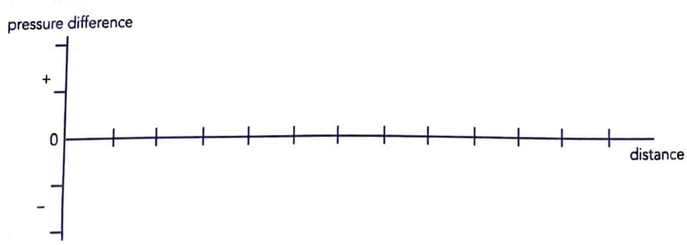
[1 mark]

(b) Complete the graph below to show how air pressure varies with distance from the microphone 4.00 ms after the commencement of the sound. Assume that initially a compression wave leaves the cone. Pressure levels are arbitrary. Select appropriate distance units.

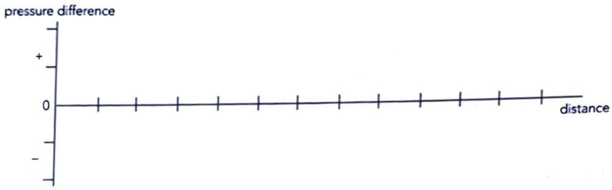


[4 marks]

(c) How long will it take sound to reach the microphone after the commencement of the sound?

[1 mark]

(d) Sketch a pressure difference versus time graph as would be recorded at the microphone for the first 0.010 seconds of sound emitted by the speaker.



[4 marks]

7. Paula and Melissa carried out an experiment using a sound level meter to take readings of the sound from a CD player. Their results for different distances are shown below.

DISTANCE FROM CD PLAYER (m)	2	4	6	8	10	12
SOUND INTENSITY (Wm-2)	1.0×10^{-3}	2.5×10^{-4}	1.12×10^{-4}	6.31×10^{-5}	4.0×10^{-5}	1.58×10^{-5}

(a) By what factor did the intensity of the sound change between:

(i) 2 m and 4 m

(ii) 4 m and 8 m

[3 marks]

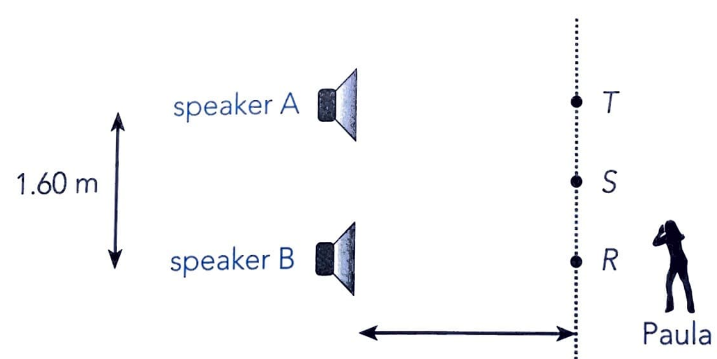
(b) Use the data to construct a graph of sound intensity versus distance from the source.

[4 marks]

(c) Using the graph (or data) determine the relationship between sound intensity and the distance of a sound from its source.

[3 marks]

8. Paula and Jake connect two similar speakers (A and B) to a sound frequency generator so that each speaker will emit sounds that are in phase and of equal frequency and intensity. They investigate the intensity of the sound produced at points along a line parallel to the speakers. They use a frequency of 685 Hz. Dimensions of the layout of the experiment are shown below.



Paula walks along the line from R to T and notices that maximum sound intensities occur at R and T (directly in front of the speakers) and at S (a point equidistant from each speaker). Quiet spots are noticed in between.

(a) What is the cause of this effect? Explain clearly.

[2 marks]

(b) From the diagram determine the difference in the distance between AT and BT.

[3 marks]

(c) Use your result in (b) to determine the velocity of sound during this experiment.

[3 marks]

(d) Paula and Jake decide to double the frequency of the sounds from each of the speakers while keeping the intensities equal. Describe the difference that Paula will notice as she again walks from R to T.

[2 marks]

SECTION THREE – COMPREHENSION (12 MARKS)

Locating earthquakes

The actual location of a seismic event such as an earthquake can be found if the seismic recordings from at least three recording stations are combined and analysed. Seismic waves are caused by events such as earthquakes and they travel through the earth's crust in different forms and varying speeds. Seismic waves travel through the earth's crust both as longitudinal and transverse waves.

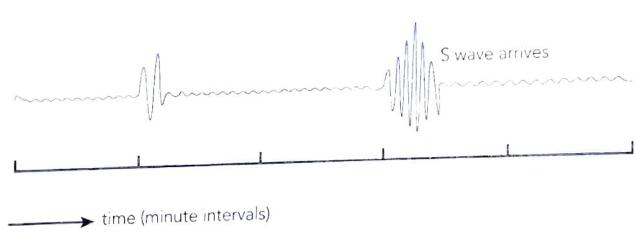
The longitudinal waves travel almost twice as fast as the transverse waves and are the first to arrive after a seismic event. They are referred to as *P waves* (primary). They tend to be heard rather than felt. Their velocity in the earth's crust can range between 4.0 km/sec to 8.0 km/sec., in water it is 1.5 km/sec and in air, the usual speed of sound, about 0.33 km/sec.

The transverse waves travel more slowly and are second to be felt. They are called *S waves* (secondary). These waves have more energy, move at right angles to their direction of travel and are generally more destructive. Their velocity in the earth's crust can range between 2.5 km/sec to 4.0 km/sec. S waves do not travel through liquids or air.

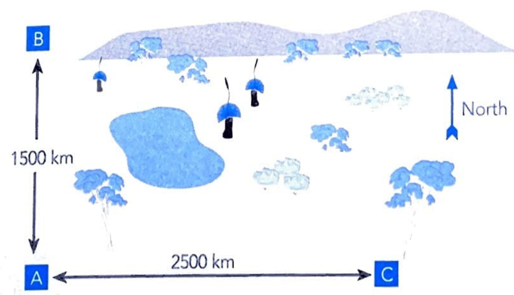
The difference in time of arrival between P and S waves provides a simple and direct way of calculating the distance of a seismic event. Conversion tables have been established by analysing data from many earthquakes and these give distances of about 9.0 km for every second of time difference. To find the actual location an earthquake event the first step is to find its distance from three different recording stations. These distances can then be plotted to scale as circles around each recording station. The epicentre is where the circles intersect.

The seismic recording from a recording station A is shown below. Similar recording were also made at two other stations B and C, located 1500 km due North and 2500 km due East of recording station A. Their relative positions are also shown below. Use all the information and data given to answer the following questions.

Seismogram – Recording station A



Seismic Stations



9. (a) Why do you think P waves (primary waves) are so called?

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[2 marks]

9. (b) P waves tend to be both felt and heard while S waves can only be felt. Explain the likely reason for this difference.

[2 marks]

(c) A seismogram for the arrival of P and S waves at a recording station A is shown above. Use it to determine the time difference of arrival of the P and S waves at recording station A.

[2 marks]

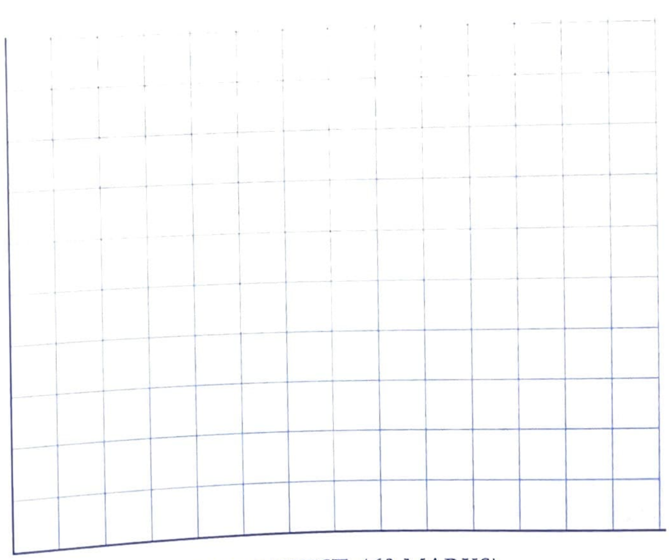
(d) Assuming, as given above, a value of 9.0 km for every second of S-P time difference, calculate the distance of the seismic event from station A.

[2 marks]

(e) Similar analysis of the recordings at stations B and C establish that they are 1200 km and 1750 km from the seismic event. The relative locations of stations A, B and C are given above.

Use the graph grid below to show these locations to scale. Then, by also drawing circles to scale, representing the distances of each station from the seismic event, determine the approximate location of the epicentre.

[4 marks]



END OF TEST (60 MARKS)